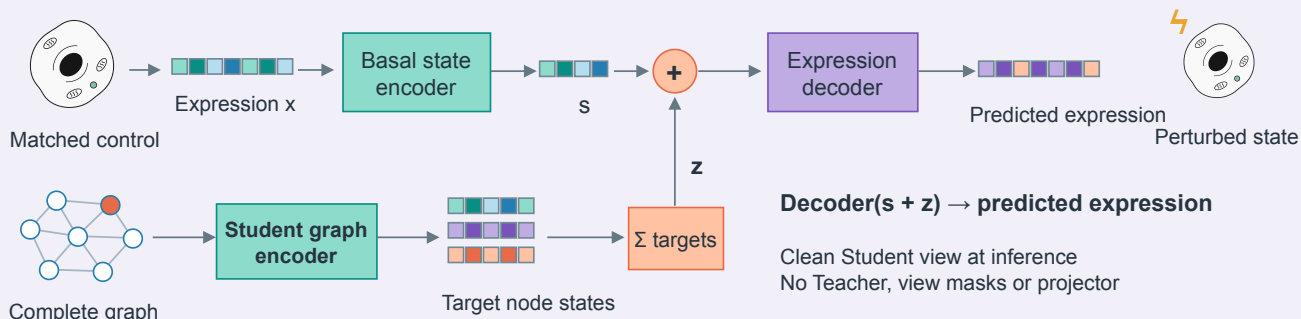


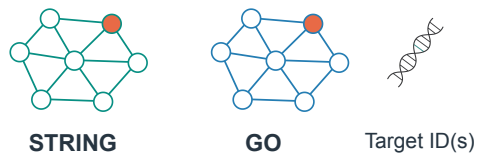
Overview

Joint perturbation prediction and graph self-distillation

a Control-conditioned prediction



b Biological priors and graph views



Gene nodes • source-specific relations



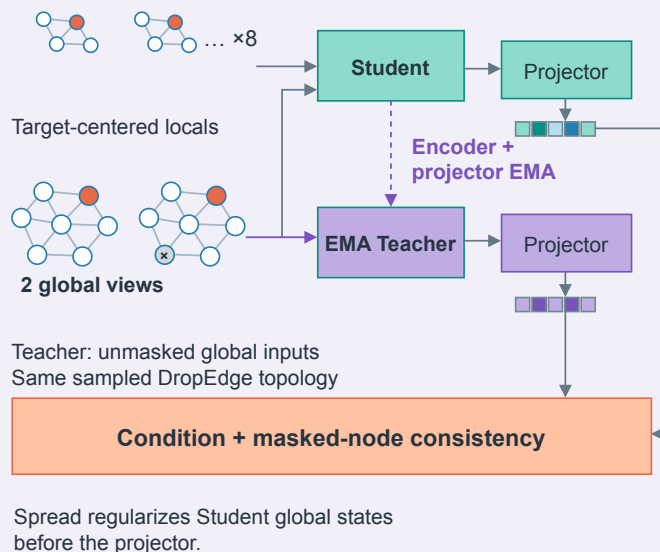
Global context Target-centered locals

2 global views • 8 local views

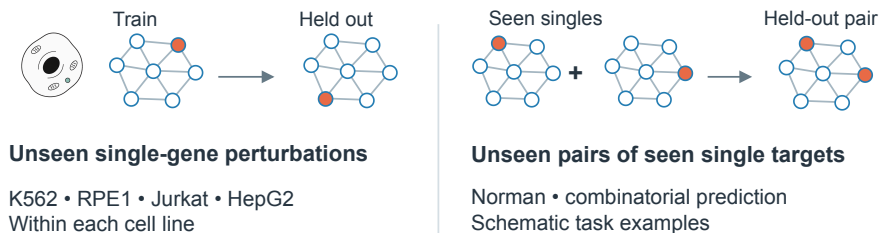
Local anchors retained; global node masking

● Perturbed gene ○ Context gene

c Learning across graph views



d Broader five-dataset benchmark and joint objective



Prediction + self-distillation

$L = L_{pred} + 0.8 L_{cond} + 0.4 L_{mask} + 0.1 L_{spread}$
 L_{pred} : MSE against observed perturbed expression